

KING SALMON, AK, USA

WMO: 703260

Lat: 58.683N

Lon: 156.656W

Elev: 20

StdP: 101.08

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25503

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.2	-27.6	-33.5	0.2	-29.8	-30.2	0.2	-26.7	15.9	3.7	13.8	3.2	1.9	10	0.769

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	22.9	15.3	20.8	14.4	18.9	13.4	16.2	21.2	15.2	19.3	14.2	17.6	4.2	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.1	10.1	17.4	13.3	9.5	16.5	12.4	9.0	15.5	45.3	21.2	42.3	19.3	39.7	17.7	23.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.4	10.8	9.1	-33.0	26.1	4.5	2.5	-36.3	27.9	-38.9	29.4	-41.4	30.8	-44.7	32.7		
			-32.5	17.9	3.6	1.9	-35.1	19.2	-37.2	20.4	-39.2	21.4	-41.9	22.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.2	-8.6	-5.1	-5.2	1.7	7.1	11.2	13.3	12.8	9.1	2.2	-4.2	-7.8
	DBStd	10.04	10.20	8.28	7.36	4.41	3.06	2.51	2.38	2.35	3.06	5.30	7.33	9.40
	HDD10.0	3107	576	422	471	248	97	14	2	4	51	243	426	553
	HDD18.3	5872	834	655	729	498	347	215	158	171	277	500	676	812
	CDD10.0	278	0	0	0	0	8	49	104	92	23	2	0	0
	CDD18.3	2	0	0	0	0	0	0	1	1	0	0	0	0
	CDH23.3	47	0	0	0	0	3	13	24	7	0	0	0	0
	CDH26.7	4	0	0	0	0	0	1	3	1	0	0	0	0
Wind	WSAvg	4.1	4.4	4.5	4.3	4.4	4.3	3.7	3.7	3.7	3.9	4.1	4.1	4.1
Precipitation	PrecAvg	521	26	21	23	26	36	43	59	80	77	55	40	35
	PrecMax	833	61	57	87	76	85	83	126	145	185	118	123	93
	PrecMin	336	4	2	0	5	6	5	19	27	23	1	0	4
	PrecStd	93	14	15	19	16	18	19	22	28	37	25	24	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	7.4	9.1	14.6	22.4	25.7	26.2	24.1	19.2	13.8	10.1	8.0
		MCWB	5.1	4.4	4.9	7.9	12.2	15.2	17.6	16.7	13.8	9.9	7.6	5.1
	2%	DB	5.8	5.9	6.9	11.8	18.8	22.1	23.0	21.3	16.9	12.2	8.5	6.2
		MCWB	3.7	3.4	3.9	6.3	10.6	14.0	16.1	15.3	12.3	9.1	6.3	4.0
	5%	DB	4.8	4.9	5.3	10.1	15.8	19.6	20.7	19.4	15.3	10.8	6.9	5.0
		MCWB	3.0	2.8	2.8	5.5	9.2	13.0	14.9	14.5	11.6	8.2	5.2	3.2
	10%	DB	3.8	3.9	4.0	8.5	13.5	17.3	18.6	17.7	14.0	9.5	5.2	3.8
		MCWB	2.3	2.0	1.9	4.7	8.1	11.8	13.8	13.6	11.1	7.2	3.7	2.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.3	5.0	5.4	8.8	13.2	16.2	18.3	17.8	15.0	11.0	8.0	5.6
		MCDB	7.4	6.7	8.3	13.5	21.3	23.5	24.9	22.7	17.5	12.8	9.5	7.4
	2%	WB	4.0	3.8	4.1	6.8	11.0	14.6	16.6	16.0	13.1	9.7	6.7	4.2
		MCDB	5.5	5.5	6.5	11.0	17.6	20.9	22.2	19.9	15.5	11.6	8.2	5.8
	5%	WB	3.1	3.0	3.0	5.7	9.7	13.4	15.3	15.1	12.1	8.5	5.3	3.3
		MCDB	4.6	4.7	5.1	9.6	15.1	18.6	19.6	18.4	14.4	10.4	6.9	4.8
	10%	WB	2.3	2.1	2.0	4.9	8.6	12.2	14.3	14.1	11.4	7.6	3.8	2.4
		MCDB	3.7	3.6	3.7	8.2	12.9	16.6	17.8	16.9	13.5	9.5	5.1	3.8
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.0	9.2	9.3	10.3	10.2	8.5	8.3	8.4	8.1	7.6	7.9
		MCDBR	5.3	6.7	8.4	12.0	16.3	15.7	13.7	12.3	10.5	7.8	6.5	6.1
		MCWBR	4.1	5.2	6.1	7.6	9.0	8.4	7.4	6.9	6.5	5.5	5.3	4.8
	5% WB	MCDBR	5.3	6.5	7.7	10.9	14.0	13.9	12.5	10.4	8.6	6.9	6.6	5.8
		MCWBR	4.2	5.2	5.7	7.1	8.0	7.7	7.0	6.1	5.6	5.2	5.6	4.8
Clear-Sky Solar Irradiance	taub	0.265	0.307	0.317	0.361	0.384	0.377	0.371	0.365	0.334	0.307	0.307	0.266	
	taud	2.162	2.223	2.255	2.255	2.267	2.365	2.427	2.445	2.502	2.436	2.073	2.016	
	Ebn at Noon	554	709	822	842	845	858	855	834	809	720	458	385	
	Edh at Noon	47	74	95	114	121	111	103	94	76	60	52	39	
All-Sky Solar Radiation	RadAvg	0.44	1.08	2.76	4.12	4.96	5.23	4.32	3.58	2.55	1.38	0.58	0.27	
	RadStd	0.06	0.17	0.39	0.43	0.49	0.50	0.46	0.41	0.31	0.14	0.07	0.03	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	+2.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page